1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L15 – Rapid Prototype Evaluation I**

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3 **Recap**

- We learned how to create prototypes
- Before we go for implementation, we evaluate those to refine our designs
- Since it is iterative, we need to do so fast

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5 **Recap**

- Today, we'll learn about quick prototype evaluation

6

Evaluation

7 **Basic Idea**

- Let's start with a basic question
  - How do we evaluate an interface?

8 **Basic Idea**

- Answer – by evaluating usability
- How?

9

Expert Evaluation

10 **Basic Idea**

- Used for quick and cheap evaluation (typically done at early stages of design)
- We need TWO things

11 **Basic Idea**

- At least a low-fidelity prototype
- An evaluation team
  - May be the design team or may include other skilled designers
  - Optionally, few end users may also be included
  - Should have at least 3-5 members

12 **Basic Idea**

- Each team member evaluates individually and produces a report
  - Report contains a list of usability issues
- All these reports combined to produce a final list

13 **Note**

- We'll learn TWO of the expert evaluation methods

14

Cognitive Walkthrough

15 **Basic Idea**

- An Usability Inspection Method
- Requirements
  - At-least a low-fidelity prototype with support for several tasks (vertical prototype)
  - 3-5 member team (as mentioned earlier)

16 **Example – Prototype Building**

- Let us consider the simple calendar app. In our app, we show only the months in the first screen. Once the user selects a month, another screen appears showing the dates in that month. A user can select any date and add some note. What are the tasks that a user can do with this interface?

17 **Example – Prototype Building**

- Some of the tasks
  - Select a month
  - Select a day of a month
  - Add note
  - Get back to the month view from the day view

18 **Example – Prototype Building**

- To perform cognitive walkthough, we should make prototype that supports more than one of these tasks
- We can make simple paper prototypes for these tasks
  - Each prototype series of sketches depicting change of screen after each interaction

19 **Additional Requirements – Task Description**

- Task descriptions - specify scenario to perform tasks
- Ex (calendar app): You are planning to take a lecture on human-computer interaction. You wish to schedule a class on the first Monday of the next month. The students are available only on the specific Monday. On all the other days, they do not have any free slot! You want to find out the date so that you can inform the students about the class.

20 **Additional Requirements – Task Description**

- Task - identify date on which first Monday of next month falls
- Interface-level tasks

- Select the next month
- Locate the first Monday
- Mark the date

21 **Procedure**

- The scenario is given to the evaluators
- They are asked to find out the date (by performing the interface-level tasks with the prototype)

22 **Additional Requirements – Questions**

- We also need to frame a set of questions (usability issues) beforehand
- Evaluators report on these issues while they perform the tasks

23 **Additional Requirements – Questions**

- Example questions (calendar app)
  - Are you able to locate the month you are looking for easily?
  - Is the interaction required to change from the month view to the day view apparent?
  - Did you find it difficult to locate the first Monday?
  - Was the date clearly visible along with the day?
  - Did you try to go back to the month view? Was the mechanism to go back clearly visible?

24 **Additional Requirements – Questions**

- Purpose - identify problems user likely to face
- Evaluators not expected to answer only "yes" or "no"
  - Should give a detailed report on what they felt about the interfaces and interactions with respect to each question
- Feedbacks are analyzed to identify broader usability issues

25 **Additional Requirements – Questions**

- Is the effect of the action same as that of the goal of the user at that point?
- Will a user see the control for a particular action?
- Will a user see that the control produces the desired effect?
- Will a user select a different control instead?
- Will a user understand the feedback provided by the system to proceed correctly?
- What happens in case of an error?
- How a user, who is familiar with other systems that perform similar tasks, is going to react?

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Heuristic Evaluation

27 **Basic Idea**

- Cognitive walkthough useful in the early stages of design
- Problem - method is scenario-based
  - we evaluate w.r.t. specific usage scenarios
  - Complex systems likely to be numerous usage scenarios

- Can't evaluate w.r.t. all

28 **Basic Idea**

- TWO reasons
  - First, we may not have that much time
  - Second (and more likely), we may not even be able to enumerate all possible usage scenarios

29 **Basic Idea**

- We can work with representative use cases
  - Not easy either

30 **Basic Idea**

- We may go for comprehensive evaluation of the whole system
  - We no longer need to create scenarios and ask evaluators to perform tasks
  - Instead, we can ask evaluators to tick on a checklist of features of the system as a whole

31 **Basic Idea**

- That is heuristic evaluation – evaluate a system with a checklist
  - Items in the checklist are called heuristics

32 **Basic Idea**

- Many such checklists available
  - Some are quite detailed and system specific
  - There are some that are more focused on broader principles of usability
- Latter category contains fewer heuristics

33 **Basic Idea**

- We shall discuss one from this latter category, which happens to be very popular as well – the 10 heuristics by Nielsen [Nielsen, 1994]

34 **Nielsen's 10 Heuristics**

35 **Setup & Procedure**

- Low-fidelity prototypes - horizontal prototypes alright
  - Unlike the walkthrough method, which requires vertical prototypes

36 **Setup & Procedure**

- We also need a team of evaluators having similar team size (at least 3-5 members)
- Can be designers or other expert designers

37 **Setup & Procedure**

- Evaluation process slightly different
  - Task scenario not required - each evaluator checks design w.r.t. heuristics and report their findings
  - Reports are combined to determine heuristics that are violated

38 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India

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Chapter 9, Sec 9.1-9.2